

RITVICK RAGHUVANSHI

☎ (209)-637-541 ✉ ritvick_raghuvanshi@ucsb.edu 🔗 linkedin.com/in/rraghuvanshi1 🐙 github.com/ritvick4u

Education

University of California, Santa Barbara

Expected June 2029

B.S. Mechanical Engineering, Current Sophomore, Standing Junior, Current GPA: 3.88/4.00

Santa Barbara, CA

- **Relevant Coursework:** Machine Shop, Graphics, CAD & Design, Statics, Strength of Materials
- **Planned Coursework by Summer 2027:** Dynamics, Thermodynamics, Fluid Mechanics

Experience

Gaucha Racing (Formula SAE)

2025 – Present

Drivetrain Division Member

University of California, Santa Barbara

- Drew free body diagrams and identified loading conditions for the drivetrain mounting bracket design
- Designed mounting brackets in **SolidWorks** and performed **FEA** to validate strength before manufacturing
- Iterated bracket geometry from FEA results, and selected final material based on strength-to-weight
- Created manufacturing drawings with **GD&T**, performed tolerance analysis, and sent documentation to machine shops

Neurotech

Jan. 2026 – Present

Mechanical Designer, NeuroDrone Project

Santa Barbara, CA

- Designing the main chassis, arms, and landing legs in CAD for a custom-built drone as part of an EMG-control research project
- Designing housings to mount and protect the onboard Arduino, motors, and **ESP32** microcontroller, and ESP32 Camera
- Integrating flight electronics and wiring with the mechanical structure, iterating geometry as new sensors/components are added
- Currently collaborating with a team to developing EMG-based flight control

VEX Robotics – UC Santa Barbara (VEX U)

Sept. 2025 – Present

Mechanical Designer

Santa Barbara, CA

- **Mechanical Designer** for UCSB VEX U team, I designed **2** VEXU competition robots to date in **Fusion 360**
- Designed drivetrain and scoring subsystems, and produced engineering drawings and BOMs each build cycle
- Manufactured components via **3D printing**, assembled and tested robots for competition reliability
- Calculated gear ratios and iterated subsystem geometry between competitions based on match testing
- Qualified for **VEX World Championship**, placing **4th** in division
- Also competed **4** years in VEX V5RC at the high school level (2021 – 2025) as **Team Captain & Mechanical Lead**, designing and building **4** V5RC competition robots

Personal Projects

Smart Bins | *Mechanical Sorting System (SolidWorks, Arduino, Raspberry Pi)*

- Designed a mechanical sorting mechanism using **6 servos** and **4 ultrasonic sensors** for automated waste separation
- Applied **Hooke's Law** for dynamic mass estimation; integrated servos and sensors with Arduino/Raspberry Pi control
- Rapid-prototyped, assembled, and tested a fully functional unit in under **36 hours** at a hackathon

Robotic Hand | *Cable-Driven Hand (SolidWorks, 3D Printing)*

- Designed **5** independently actuated fingers driven by a cable-tendon system and **5** servo motors
- Designed finger joints and iterated geometry across multiple prototype revisions
- 3D printed, assembled, and validated a functional prototype

Boy Scouts Eagle Project | *Autism Center Balance Beam Construction*

- Designed and built **3** balance beams for motor-skill development at a local autism center
- Used CAD modeling to size structural members for durability and ergonomics
- Led **25** volunteers through fabrication and assembly; verified structural safety and managed budgeting to finish under **\$1,000**

Technical Skills

- **CAD & Design:** SolidWorks, Fusion 360, CREO, Engineering Drawings, GD&T, Tolerance Analysis, BOM Creation
- **Analysis:** SolidWorks Simulation (FEA), Engineering Calculations, MATLAB
- **Manufacturing:** 3D Printing, Fusion CAM, CNC Machining, Mill, Lathe, Rapid Prototyping
- **Programming:** Python, MATLAB, Java, C++, Arduino, Raspberry Pi

Awards

- VEX World Championship Qualifier | 4th Place Division Finish | VEX Create Awards | VEX Design Award | Eagle Scout